

Meningococcal Prophylaxis & Viral Meningitis — Drugs and Discriminators

MENINGOCOCCAL PROPHYLAXIS STATION

DEFINITION BOARD

FOR CLOSE CONTACTS

KISSING

SHARING BEVERAGES

HOUSEHOLD MEMBERS

DORM ROOMMATES

HEALTHCARE WORKER
— INTUBATION WITHOUT PPE
— NEEDS PROPHYLAXIS

DOCTOR IS NOT EXEMPT

INDEX CASE
— GETS PROPHYLAXIS TOO
— ERADICATE CARRIAGE

DRUG DISPENSARY COUNTER

RIFAMPIN
— ADULTS 600 mg PO q12h x 2 DAYS
— ~100% EFFECTIVE
— FIRST-LINE ORAL

NOT IN PREGNANCY

CEFTRIAXONE IM
— ADULTS 250 mg x 1 SINGLE DOSE
— PREFERRED IN PREGNANCY

AZITHROMYCIN
500 mg PO x 1 DOSE
ALTERNATIVE

CIPROFLOXACIN
500 mg PO x 1 DOSE
ADULTS ONLY

PENICILLIN TRAP
PENICILLIN = DOES NOT ERADICATE NASOPHARYNGEAL CARRIAGE
AMOXICILLIN — SAME PROBLEM
= INEFFECTIVE FOR PROPHYLAXIS

TIMING MYTH BUSTER
PROPHYLAXIS NOT LIMITED TO FIRST 4-8 HOURS
— EFFECTIVE BEYOND THIS WINDOW

VIRAL MENINGITIS CLINIC

WAITING ROOM

VIRAL SUSPECTS IN ORDER OF FREQUENCY

ENTEROVIRUSES (COXSACKIE + ECHOVIRUS)
— >85% OF VIRAL CAUSES
— MOST COMMON
— SUMMER/FALL
— CHILDREN <15
— DIAGNOSE BY CSF RT-PCR
— >95% SENSITIVE

1
HAND-FOOT-MOUTH
MYOPERICARDITIS

HSV-2
— UNCOMPLICATED MENINGITIS
— 5% ADULTS ~ 20%
— RECURRENT ATTACKS (MOLLARET'S)
— TREAT WITH ACYCLOVIR

2
MANY HAVE NO ACTIVE LESIONS AT PRESENTATION

VZV
— 3-20% VARIABLE
— CAN OCCUR WITHOUT RASH
— DIAGNOSE PCR + INTRATHECAL VZV ANTIBODY

3

WEST NILE VIRUS
— SUMMER/FALL
— 45% PMN
— PLEOCYTOSIS
— GLUCOSE NORMAL DESPITE PMNs
— DIAGNOSE CSF IgM >80%
— PCR ONLY 70%

4
PARADOX

CMV
— LEAST COMMON FROM USUAL OPTIONS
— MAINLY IMMUNOCOMPROMISED
— HIV/TRANSPLANT

5

CRITICAL DRUG DISTINCTION

ACYCLOVIR = HSV + VZV
GANCICLOVIR = CMV ONLY
NEVER INTERCHANGE
Q15 AND Q17 BOARD TRAP

FALSE STATEMENTS BOARD

LOW CSF GLUCOSE = TYPICAL OF VIRAL
PMN PREDOMINANCE = TYPICAL OF VIRAL
STUPOR + COMA = VIRAL MENINGITIS (= ENCEPHALITIS)
FOCAL SIGNS = VIRAL MENINGITIS (= ENCEPHALITIS)
— ADD ACYCLOVIR

HSV-1 ENCEPHALITIS NOTE
FOCAL NEUROLOGICAL SIGNS + FEVER
= ADD ACYCLOVIR EMPIRICALLY
HSV-1 ENCEPHALITIS IS #1 MIMIC OF BACTERIAL MENINGITIS WITH FOCAL FEATURES

1. Close-contact definition

WHAT YOU SEE

Board listing kissing, shared drinks, household, dorm roommates

WHAT IT ENCODES

Meningococcal prophylaxis is given to close contacts: household members, kissing/shared utensils, dorm/military barracks roommates, and anyone with direct exposure to oral secretions within 7 days before onset. Casual classroom or workplace contact does NOT qualify.

BOARD TRAP

Healthcare workers are NOT routine close contacts — only those with intimate airway/secretion exposure (intubation, mouth-to-mouth, suctioning without a mask) need prophylaxis.

2. HCW exception

WHAT YOU SEE

Healthcare worker panel marked 'not exempt' only if airway exposed

WHAT IT ENCODES

Most HCWs caring for a meningococcal patient need no prophylaxis. Only those who performed unprotected mouth-to-nose contact (intubation, airway suctioning, mouth-to-mouth) before 24h of effective antibiotics qualify.

BOARD TRAP

Wrong answer: give prophylaxis to the whole ward staff. Discriminator = documented unprotected secretion/airway exposure.

3. Index case treats carriage

WHAT YOU SEE

Index case at counter getting prophylaxis to eradicate carriage

WHAT IT ENCODES

The index patient also needs carriage-eradication therapy unless treated with ceftriaxone, because IV penicillin does NOT reliably clear nasopharyngeal carriage. Add rifampin or a single ceftriaxone/cipro dose before discharge.

BOARD TRAP

Wrong: penicillin-treated index patient is non-infectious — penicillin does not eradicate carriage, so they can still transmit.

4. Rifampin

WHAT YOU SEE

Rifampin vial, adults 600 mg q12h x2 days

WHAT IT ENCODES

Rifampin 600 mg PO q12h x 4 doses (2 days) is first-line prophylaxis. Avoid in pregnancy and warn about drug interactions (induces CYP450) and orange body fluids. ~90% carriage eradication.

BOARD TRAP

Rifampin is NOT first-line in pregnancy — use ceftriaxone instead.

5. Ceftriaxone (pregnancy choice)

WHAT YOU SEE

Ceftriaxone syringe, 250 mg IM single dose

WHAT IT ENCODES

Ceftriaxone 250 mg IM (125 mg in children) as a single dose is the preferred prophylaxis in pregnancy and a reliable one-dose option. Single IM dose aids compliance.

BOARD TRAP

Wrong: oral agent for a pregnant contact — single-dose IM ceftriaxone is the safest pick.

6. Ciprofloxacin

WHAT YOU SEE

Cipro tablet, 500 mg single oral dose

WHAT IT ENCODES

Ciprofloxacin 500 mg PO once is a convenient single-dose adult option. Avoid in pregnancy and in regions with documented fluoroquinolone-resistant N. meningitidis.

BOARD TRAP

Cipro single dose is fine for adults but NOT for pregnant contacts.

7. Azithromycin alternative

WHAT YOU SEE

Azithromycin, 500 mg single dose alternative

WHAT IT ENCODES

Azithromycin 500 mg PO once is an alternative when fluoroquinolone resistance is a concern; it is also acceptable in pregnancy.

BOARD TRAP

Not first-line broadly, but the go-to when cipro resistance is documented.

8. Penicillin trap

WHAT YOU SEE

Penicillin marked 'does NOT eradicate carriage'

WHAT IT ENCODES

IV penicillin treats meningococcal disease but does not clear nasopharyngeal carriage, so the treated patient still needs carriage eradication. Amoxicillin shares this limitation.

BOARD TRAP

Classic board trap: penicillin = treated = no carriage. False — must still eradicate.

9. Timing window

WHAT YOU SEE

Prophylaxis effective if given days 1–7, not just immediately

WHAT IT ENCODES

Prophylaxis is effective when given up to ~7–10 days after exposure, not only in the first hours. Late presentation within the window still warrants treatment.

BOARD TRAP

Wrong: prophylaxis is useless if not given immediately. It remains effective across the 7-day window.

10. Enterovirus (top viral cause)

WHAT YOU SEE

Enterovirus panel, most common, summer/fall, hand-foot-mouth

WHAT IT ENCODES

Enteroviruses (coxsackie, echovirus) cause >85% of viral (aseptic) meningitis, peak in summer/fall, often with hand-foot-mouth or myopericarditis. CSF shows lymphocytic pleocytosis with normal glucose; supportive care only.

BOARD TRAP

No antiviral for enteroviral meningitis — treatment is supportive.

11. HSV-2 → treat with acyclovir

WHAT YOU SEE

HSV-2 panel, recurrent meningitis (Mollaret), treat with acyclovir

WHAT IT ENCODES

HSV-2 causes recurrent benign lymphocytic (Mollaret) meningitis and is the classic viral meningitis you DO treat — IV acyclovir. PCR of CSF is diagnostic.

BOARD TRAP

HSV-2 = meningitis treated with ACYCLOVIR; do not confuse with CMV (needs ganciclovir).

12. VZV meningitis

WHAT YOU SEE

VZV panel, can occur without rash, intrathecal antibody

WHAT IT ENCODES

VZV can cause meningitis/encephalitis even without a vesicular rash; treat with acyclovir. Diagnose by CSF PCR or intrathecal anti-VZV antibody.

BOARD TRAP

Absence of rash does not exclude VZV — still treat with acyclovir.

13. CMV — ganciclovir, immunocompromised

WHAT YOU SEE

CMV panel, common in HIV/transplant

WHAT IT ENCODES

CMV CNS disease occurs mainly in immunocompromised (HIV CD4<50, transplant) patients and is treated with ganciclovir/foscarnet, NOT acyclovir.

BOARD TRAP

Acyclovir does NOT cover CMV — the discriminator is ganciclovir for CMV.

14. Critical drug distinction box

WHAT YOU SEE

Red box: acyclovir = HSV/VZV; ganciclovir = CMV; never acyclovir for CMV

WHAT IT ENCODES

High-yield Q15/Q102 trap: Acyclovir covers HSV-1/2 and VZV; ganciclovir covers CMV. Acyclovir never works for CMV. Match the virus to the right antiviral.

BOARD TRAP

Single most common viral-meningitis board trap — picking acyclovir for a CMV/immunocompromised case.

15. Encephalitis vs meningitis

WHAT YOU SEE

False-statements board: focal signs/altered mentation = encephalitis

WHAT IT ENCODES

Pure viral meningitis = low CSF glucose is FALSE (glucose is normal); preserved mentation. Stupor, focal neuro signs, or seizures shift the diagnosis to encephalitis (think HSV-1 temporal lobe).

BOARD TRAP

Focal signs + fever = encephalitis (HSV-1), not simple viral meningitis — and HSV encephalitis is a treat-now emergency.
